

APPENDIX A

Subject Handbook

This appendix consists of sample subject handbooks for the choice experiments in which subjects did not engage in production. Two of the five separate treatments are reflected here. One is the basic experiment, in which subjects stood to gain income and in which the exemplary income distributions did not have very large variances. The actual document consists of all text not in boldface. The second handbook presented is for the variation of the basic experiment in which all reference to justice was removed. It can be read by deleting all regular type in square brackets ([]) and replacing it with all **boldface** type in braces ({}). The two handbooks are identical except for the references to fairness and justice contained in the basic experiment but not in the variation.

Both handbooks include descriptions of the principles for redistributing payoffs, sample income distributions, sample problems, tests, four choice situations that provide subjects with experience in choosing principles and getting a random assignment to a payoff class, several measures of preferences for principles, and rules for discussion and decision making. This handbook does not contain the debriefing questionnaire.

This experiment deals with the [question: "What is a just distribution of income?" An individual's lifetime income is in

part a result of many genetic and social accidents. The luckiest get the greatest talents and the highest rewards such as status and wealth. The least fortunate get the lowest abilities and opportunities, and receive the associated costs of poverty. Societies can deal with these inequities and risks by adopting income redistribution policies. This experiment deals with the justice of such policies.] **{question of the distribution of monetary gains or losses among members of committees.}** The experiment is divided into three parts.

In the first part of the experiment each of you will be introduced to a few [theories of justice.] **{possible rules for distributing the monetary gains and losses among committee members.}** To do this you will consider some examples and make some choices. These choices will have real monetary consequences for you. Your pay for the first part of the experiment will be based on your choices. You will have 1 hour for the first part. In this part you will be given a series of questions. These questions are merely to ensure that you have learned the concepts which are being used in the experiment. If you do not answer the questions correctly, then you *are to* go back to review the material and correct wrong answers. Once you have mastered the material, you can go on to make choices for which you will be paid. If you do not learn the material in a reasonable amount of time, you will not be able to earn as much money as possible since you must finish the first part of the experiment in 1 hour. But you *should* have plenty of time to finish this part of the experiment. Everyone will go on to the second part either after 1 hour or after everyone has finished Part I, whichever occurs first.

In the second part, you will all be asked, as a group, to discuss [notions of justice. After the discussion, you will be asked to reach a group decision on which principle of justice you like best. Your pay for Part II of the experiment will be based on the principle which the group chooses.] **{possible rules for distributing monetary gains or losses. After the discussion, you will be asked to reach a group decision on which rule for distributing the money you like best. Your pay for Part II of the experiment will be based on the rule which the group chooses.}**

In the third part, you will be asked some background questions about yourself. Upon completion of the third part you will receive the sum of your earnings from the two parts of the experiment by check. The money you receive is to be yours alone. No discussion or agreement to share earnings is permitted.

Throughout the experiment, we shall scale all examples and choices so that the monies can be thought of as average lifetime incomes. We then refer to these stakes as incomes. In Part I your actual stakes are equal to \$1 for every \$10,000 of income listed in the text.

PART I

[This experiment is concerned with the justice of different income distributions. Let us begin by discussing some ways of judging the justice of an income distribution.] **{This experiment is concerned with committee decisions about rules to distribute income among members of a society which *includes* you. Let us begin by discussing the effect of different rules on what you will get in the experiment.}** One way of judging [the justice of an income distribution] **{the effect of different rules on what you will get}** is to consider the overall pattern of income distribution. A number of [principles] **{rules}** have been suggested for [these sorts of judgments] **{adoption for the distribution of income}**, and we shall illustrate them by pointing out four such [principles.] **{rules.}** But there are obviously other possible alternatives. *You may well think of some alternatives yourself!*

To illustrate such principles consider the following four notions [of justice]:

1. Maximizing the Floor Income

[The most just distribution of income is that which maximizes the floor (or lowest) income in the society.] **{The best rule for distributing income is that which maximizes the floor (or lowest) income in the society.}**

This [principle] **{rule}** considers only the [welfare] **{income}** of the worst-off individual in society. In judging among income distributions, the distribution which ensures the poorest person **{who might be you}** the highest [income is the most just. No] **{income, ensures that no}** person's income can go up unless it increases the income of the people at the very bottom.

2. Maximizing the Average Income

[The most just distribution of income] **{The best rule for distributing incomes}** is that which maximizes the average income in the society.

For any society maximizing the average income maximizes the total income in the society. Since you don't know what you may get, this maximizes your average or mathematically expected income.

3. Maximizing the Average with a Floor Constraint of \$——

[The most just distribution of] **{The best rule for distributing income}** is that which maximizes the average income only after a certain specified minimum income is guaranteed to everyone.

Such a [principle] **{rule}** insures that the attempt to maximize the average is constrained so as to ensure that individuals "at the bottom" receive a specified minimum. To choose this [principle] **{rule}** one must specify the value of the floor (lowest income).

4. Maximizing the Average with a Range Constraint of \$——

[The most just distribution of] **{The best rule for distributing income}** is that which attempts to maximize the average income only after guaranteeing that the difference between [the poorest and the richest individuals'] **{smallest and the largest}** incomes (i.e., the range of income) in the society is not greater than a specified amount.

Such a [principle] **{rule}** ensures that the attempt to maximize the average does not allow income differences [between rich and poor] to exceed a specified amount. To choose this [principle] **{rule}** one must specify the dollar difference between the high and low incomes.

Of course, there *are* other possible [principles,] **{rules,}** and you may think of some of them.

Now that you have been familiarized with these four [principles,] **{rules,}** please answer the following questions:

Rank order, according to your preferences, the following 4 [principles of distributive justice] **{rules for distributing income}** by placing the letters (a), (b), (c), (d), signifying the [principles] **{rules}**, in the blanks below. Indicate ties by placing the tied [principles] **{rules}** in the same space.

most preferred _____ least preferred

- a. maximize the floor income
- b. maximize the average income
- c. maximize the average income, subject to a floor constraint
- d. maximize the average income, subject to a range constraint

How do you feel about your ranking of these [principles?] **{rules?}** (Circle the appropriate answer.)

1. very unsure
2. unsure
3. no opinion
4. sure
5. very sure

To familiarize you with the sorts of choices you will be making, consider the following sample situation.

Sample Situation

Consider the following four income distributions (each of the money entries represents a yearly dollar income for a household):

Income Distributions

<i>Income Class</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
High	\$32,000	\$28,000	\$31,000	\$21,000
Medium high	27,000	22,000	24,000	20,000
Medium	24,000	20,000	21,000	19,000
Medium low	13,000	17,000	16,000	16,000
Low	12,000	13,000	14,000	15,000
Average income				
Floor or low income	12,000	13,000	14,000	15,000
Range	20,000	15,000	17,000	6,000

^a The average income was not given to the subjects at this point. It can be calculated only with knowledge of how many people will receive each level of income. We did not give the subjects any information about this.

You are to make a choice from among the four [principles of justice] **{rules}** which are mentioned above: (a) maximizing the floor, (b) maximizing the average, (c) maximizing the average with a floor constraint, and (d) maximizing the average with a range constraint. *If you choose (c) or (d), you will have to tell us what that floor or range constraint is before you can be said to have made a well defined choice.* Your choice of a [principle] **{rule}** will [select one of the four income distributions as the most just.] **{be applied to the income distributions above.}**

Only one of the income distributions is consistent with each rule. That means that your selection of a rule also selects one of the four income distributions as the one you prefer. Indicate your choice of a [principle] **{rule}** here:_____.

In making this choice, recall that your choice will yield you a payoff. How will your choice determine your pay? If, for example, you choose (a) to maximize the floor or low income, you will have picked that distribution (from the four in the sample

question) which has the highest floor. (In this case distribution 4.) This ensures that you would get at least \$15,000 if you were to be the worst off individual. This is the most that a member of the low income class could get from any of these distributions.

Let us say you choose a [principle] {rule} which leads to the selection of distribution 4: maximizing the floor or low income. After your choice has been made, five predetermined numbers will be used to determine your payoff. They represent the probability or chance that you will "be in" each of the five income classes and hence receive the payoffs associated with that income class. The numbers add up to one, and are each some fraction bigger than zero and smaller than one. Thus, the numbers can be thought of as the proportion of the population (of some hypothetical society) which receives that particular income level in the distribution. You will be assigned to an income class using these proportions. *You will then receive one dollar for each \$10,000 received by a member of the income class you have been assigned to.*

To see how your payoff is calculated, again consider the illustration above and assume you have chosen to maximize the floor income. Suppose the distribution of (a fictitious) population over these income classes has been set at:

<i>Income Class</i>	<i>Proportion of Population</i>
high	5
medium high	10
medium	50
medium low	25
low	10

This means that you have a 5% chance of being assigned to the highest income class. Were you to be so assigned, you

would get a payoff of \$2.10—the highest in the distribution #4 (in this case, the distribution which maximizes the floor or low income). Similarly you have a 10% chance of being assigned to the medium high class, etc. If you were assigned to the medium class (in this example, the most likely, with a likelihood of 50%) you would get \$1.90. *These assignments to income classes will be made randomly* by your drawing a piece of paper from a bag. You may think of them as “accidents of birth.”

Now, what if you chose to maximize the average income? The average income in the society must be calculated by taking into account the proportion of the population which receives each particular income in the income distribution. Taking this into account, the average income for each of the income distributions is as stated in the following table:

<i>Income Class</i>	<i>Income Distributions</i>			
	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
High	\$32,000	\$28,000	\$31,000	\$21,000
Medium high	27,000	22,000	24,000	20,000
Medium	24,000	20,000	21,000	19,000
Medium low	13,000	17,000	16,000	16,000
Low	12,000	13,000	14,000	15,000
Average income	20,750	19,150	19,850	18,050
Floor or low income	12,000	13,000	14,000	15,000
Range	20,000	15,000	17,000	6,000

If you chose the [principle] **[rule]** of maximizing the average income, this would select distribution #1, which has an average income of \$20,750. Therefore, on average, people selecting this [principle] **[rule]** in this example would receive \$2.07.

In all of the subsequent situations you will be given the average income for each income distribution but will *not* be told

the proportion of population in each income class. The proportions of the society in each income class *will vary* from one situation to another.

What if one chose to maximize the average with some constraint: say a floor constraint? How would this work? If you want to maximize the average while guaranteeing that no one receives an income below, let's say, \$14,000, only those income distributions with a floor of \$14,000 or more are acceptable: in this case distributions #3 and #4. Of these, distribution #3 has a higher average income and therefore maximizes the average with a floor constraint of \$14,000. Notice that lowering the constraint to \$13,000 allows us to consider distribution #2 as well. This does not affect which distribution would be picked, however, because #3 has a higher average than #2. Lowering it still further, to \$12,000, would, on the other hand, allow distribution #1 to be considered *and selected* (since #1 has the highest average).

Finally, let us consider what happens if we choose to maximize the average income with a range constraint. The most restrictive range constraint applicable here is \$6,000, which would lead to a selection of distribution #4. If one increased the permitted difference between the high and low incomes to \$15,000, income distribution #2 would be allowable and chosen since it has the higher average income, etc.

Remember, the more demanding the constraint (i.e., the smaller the range or the higher the floor), the larger the number of income distributions which are ruled out.

When you are shown your payoff, you will also be shown what you would have gotten had you chosen any of the other three [principles.] {rules.} This will permit you to see the consequences of your choices.

Before you make any choices of [principles of justice,] {rules of income distribution,} you are required to answer some questions to test your understanding of the [principles of justice.] {rules.} You may refer to any part of the booklet in deciding on your answers. When you finish answering the questions, bring your answers to the moderator. If you make mistakes, you will be given chances to correct them—but do not erase any mis-

takes after the moderator has looked at them. The following information may help you to answer the questions:

- A. The range is the difference between the highest and lowest values.
- B. The floor is the lowest value.
- C. If one chose to maximize the average "with a floor constraint" of \$10,500, this means: one selects that income distribution with the highest average such that everyone gets at least \$10,500.
- D. If one chose to maximize the average "with a range constraint" of \$10,000, this means: one selects that income distribution with the highest average such that the difference between the highest and lowest income is no more than \$10,000.

Test

Income Distributions

<i>Income Class</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
High	\$30,000	\$19,000	\$26,000	\$29,000
Medium high	25,000	18,000	20,000	22,000
Medium	20,000	16,000	18,000	18,000
Medium low	15,000	14,000	15,000	14,000
Low	10,000	13,000	11,000	12,000
Average income	20,000	16,000	18,000	19,000
Floor or low income	_____	_____	_____	_____
Range	_____	_____	_____	_____

1. In the blank space in the distributions write in the floor or low income for each distribution.
2. In the blank spaces in the distributions above write in the range for each distribution.

3. Indicate which distribution would be singled out [as the most just] by application of each of the following [principles.] **{rules.}**
 - A. Maximization of the floor income would select distribution _____.
 - B. Maximization of the average income would select distribution _____.
 - C. Maximization of the average income with a floor constraint of \$11,500 would select distribution _____.
 - D. Maximization of the average income with a range constraint of \$16,000 would select distribution _____.
4. If I were to choose to maximize the floor income, the lowest income I could get would be _____; the highest would be _____.
5. If I were to choose to maximize the average income, the lowest income I could get would be _____; the highest would be _____.

Please answer the following question, true or false:

6. After choosing a [principle of justice] **{rule about the distribution of income}**, my payoff will be determined by the probability of being in each income class and the luck of the draw. (Circle one: true or false.)

Please bring this paper to the moderator to be corrected.

Now that you have some acquaintance with the [principles of justice,] **{possible rules for distributing income,}** and before you begin making choices in the situations below, please indicate your preferences for the [principles.] **{rules.}** Rank them from most preferred to least preferred by placing the letters (a), (b), (c), and (d), corresponding to the [principles] **{rules}**, in the blank spaces below. Indicate ties by placing tied [principles] **{rules}** in the same blanks.

most preferred _____ least preferred

- a. maximize the floor income
- b. maximize the average income
- c. maximize the average income, subject to a floor constraint
- d. maximize the average income, subject to a range constraint

How do you feel about your ranking of these [principles?] {rules?} (Circle the appropriate answer.)

1. very unsure
2. unsure
3. no opinion
4. sure
5. very sure

Choices with Payoffs

Situation A Would you now please consider the income distributions below and choose a [*principle of justice*] {**rule**} which you feel would yield the best choice of an income distribution for the society. On your tally sheet place a check mark in the column labeled Situation A opposite your choice of [principle.] {**rule.**} Be sure to fill in the other information required if you choose to maximize with a constraint.

Remember, in making this choice, you will be determining a real money payoff for yourself.

To get your payoff recorded bring your tally sheet to the moderator.

DO NOT GO TO THE NEXT SITUATION UNTIL YOU HAVE TAKEN YOUR TALLY SHEET TO THE MODERATOR.

<i>Income Class</i>	<i>Income Distributions</i>			
	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
High	\$28,000	\$35,000	\$30,000	\$25,000
Medium high	25,000	30,000	29,000	22,000
Medium	20,000	25,000	28,000	19,000
Medium low	15,000	15,000	27,000	16,000
Low	12,000	10,000	6,000	13,000
Average income	20,000	23,500	26,000	19,000
Floor or low income	12,000	10,000	6,000	13,000
Range	16,000	25,000	24,000	12,000

Situation B Would you now please consider the income distributions below and choose a [*principle of justice*] **{rule}** which you feel would yield the best choice of an income distribution for the society. On your tally sheet place a check mark in the column labeled Situation B opposite your choice of [*principle.*] **{rule.}** Be sure to fill in the other information required if you choose to maximize with a constraint.

Remember, in making this choice, you will be determining a real money payoff for yourself.

To get your payoff recorded bring your tally sheet to the moderator.

DO NOT GO TO THE NEXT SITUATION UNTIL YOU HAVE TAKEN YOUR TALLY SHEET TO THE MODERATOR.

<i>Income Class</i>	<i>Income Distributions</i>			
	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
High	\$17,000	\$30,000	\$40,000	\$26,000
Medium high	16,000	25,000	30,000	24,000
Medium	15,000	20,000	25,000	22,000
Medium low	14,000	15,000	20,000	20,000
Low	13,000	12,500	8,000	11,000
Average income	14,800	19,200	23,550	20,900
Floor or low income	13,000	12,500	8,000	11,000
Range	4,000	17,500	32,000	15,000

Situation C Would you now please consider the income distributions below and choose a [*principle of justice*] **{rule}** which you feel would yield the best choice of an income distribution for the society. On your tally sheet place a check mark in the

column labeled Situation C opposite your choice of [principle.] {rule.} Be sure to fill in the other information required if you choose to maximize with a constraint.

Remember, in making this choice, you will be determining a real money payoff for yourself.

To get your payoff recorded bring your tally sheet to the moderator.

DO NOT GO TO THE NEXT SITUATION UNTIL YOU HAVE TAKEN YOUR TALLY SHEET TO THE MODERATOR.

<i>Income Class</i>	<i>Income Distributions</i>			
	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
High	\$100,000	\$35,000	\$30,000	\$24,000
Medium high	30,000	30,000	25,000	23,000
Medium	20,000	25,000	23,000	22,000
Medium low	15,000	20,000	15,000	21,000
Low	13,000	8,000	12,000	11,000
Average income	19,500	22,350	20,000	20,710
Floor or low income	13,000	8,000	12,000	11,000
Range	87,000	27,000	18,000	13,000

Situation D Would you now please consider the income distributions below and choose a [principle of justice] {rule} which you feel would yield the best choice of an income distribution for the society. On your tally sheet place a check mark in the column labeled Situation D opposite your choice of [principle.] {rule.} Be sure to fill in the other information required if you choose to maximize with a constraint.

Remember, in making this choice, you will be determining a real money payoff for yourself.

To get your payoff recorded bring your tally sheet to the moderator.

DO NOT GO TO THE NEXT SITUATION UNTIL YOU HAVE TAKEN YOUR TALLY SHEET TO THE MODERATOR.

<i>Income Class</i>	<i>Income Distributions</i>			
	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
High	\$35,000	\$30,000	\$20,000	\$30,000
Medium high	30,000	28,000	18,000	28,000
Medium	25,000	26,000	16,000	24,000
Medium low	20,000	24,000	14,000	20,000
Low	13,000	12,000	12,000	14,000
Average income	23,550	24,500	15,500	22,700
Floor or low income	13,000	12,000	12,000	14,000
Range	22,000	18,000	8,000	16,000

Now that you have been further familiarized with the four [principles,] {rules,} please answer the following question:

Rank order according to your preferences in following 4 [principles of justice] {rules} by placing the letters (a), (b), (c), (d), corresponding to the [principles] {rules}, in the blanks below. Indicate ties by placing the tied [principles] {rules} in the same space.

most preferred _____ least preferred

- maximize the floor income
- maximize the average income
- maximize the average income, subject to a floor constraint
- maximize the average income, subject to a range constraint

How do you feel about your ranking of these [principles?] {rules?} (Circle the appropriate answer.)

1. very unsure
2. unsure
3. no opinion
4. sure
5. very sure

PART II

In this part of the experiment you, as a group, are to choose one [principle of justice] {**rule for the distribution of income**} for yourselves. This choice will determine the payoff you get in this part of the experiment. Your payoffs will be determined as follows. We have constructed a large set of payoff distribution schedules. Each schedule specifies five payoff amounts. The distributions need not resemble the distributions in Part I. THE STAKES IN THIS PART OF THE EXPERIMENT ARE MUCH HIGHER THAN IN THE FIRST PART. Your choice of [principle] {**rule**} will be used to pick out those distribution schedules which conform to your [principle.] {**rule.**} Thus, for example, if you picked the [principle] {**rule**} to maximize the average income, you would be saying that the group wants to pick out a distribution with the highest average income. If there is more than one distribution which has this maximum average income, one will be picked out for you as a group. *Each of you will then be randomly assigned an income from that distribution.* That is your payoff for Part II. Before the group votes on adopting a [principle] {**rule**} [of justice], there is to be a full and open discussion of the matter before the group. You must follow the procedures below for the discussion and voting phase.

Discussion Phase

You begin by having a group discussion about which [principle] {**rule**} you should adopt. The group can terminate this discussion anytime after 5 minutes. If after 5 minutes you feel noth-

ing more can be gained by further discussion, you are to tell this to the moderator. Participants must agree *unanimously by secret ballot* that further discussion is unnecessary for discussion to be ended. You have whatever time you need, within reason, to discuss the issue. You *are not restricted*, in any way, to the four [principles of justice] **{rules}** mentioned above. Thus, you can discuss (and later adopt) other [principles.] **{rules.}** Any one of you can introduce and begin discussion of any [principle.] **{rule.}** But not just any rule is a [principle.] **{rule.}** There are two requirements which a rule must meet to be considered a [principle of justice] **{rule}**.

First, the rule *cannot use names*. For example, a rule which gives all to one *specified* person is not permitted because it specifies a *named* person. However, a rule which gives all the income to some *unspecified* person is permitted.

Second, only those rules which *always* lead to the choice of ["just distributions"] **{some income distributions}** from any set of alternatives is [a principle.] **{allowed.}** For example, strict equality is not a [principle] **{rule}** since it may not be achievable. But equality could be approximated by a [principle] **{rule}** to "minimize the range" in the distribution of income.

Finally, there are a few requirements you should bear in mind if you wish to adopt a [principle] **{rule}** which involves a constraint. You should think of dollar figures as annual family incomes in a democratic society of moderate scarcity.

If you wish to consider a range constraint, you must specify the dollar amount.

If you wish to consider a floor constraint, you are not in a position to know what floors are achievable for everyone. Of course there is a maximum achievable floor income in the set of income distributions. This means you have two ways of specifying a floor constraint: (1) You can specify an absolute dollar amount which you wish to have as a floor constraint. If that dollar amount is above the maximum floor achievable in our set of distributions, you will be assigned the income distribution with the highest achievable floor. (2) You can specify your floor constraint as a dollar amount below the maximum floor

achievable in our distributions. This will guarantee the existence of such a distribution but you will not know what the absolute dollar amount of this floor is until after the voting has been completed.

Choice Phase

After your discussion you, as a group, are to vote to adopt a [principle of distributive justice] **{rule}**. Your voting will be according to the following procedure. The group will adopt a [principle] **{rule}** if, and only if, that [principle] **{rule}** is able to secure the *unanimous* support of the group against *all* other [principles] **{rules}** sketched above, plus any others which you have discussed. The [principles] **{rules}** are to be voted upon, two at a time. Only that [principle] **{rule}** which gets *unanimous* support in two-way contests against all other [principles] **{rules}** is actually adopted by the group. If no such proposal exists, then the group will have adopted no [principle.] **{rule.}** In that case, any member of the group can ask for extra discussion, which can be terminated at any time using the procedure described above. A new vote would follow.

Your payoffs in this section of the experiment will conform to the [principle] **{rule}** which you, as a group, adopt. If you, as a group, do not adopt any [principle] **{rule}**, then we will select one of the income distributions at random for you as a group. That choice of income distribution will conform to no particular characteristics.

Remember: Whatever income distribution is chosen, YOU WILL BE RANDOMLY PLACED IN AN INCOME CLASS IN THAT DISTRIBUTION, AND THAT DETERMINES THE MONEY YOU WILL GET.

Are there any questions? If so, please ask them now, or at any time during this part of the experiment.